From PDF to Dashboard: Extracting Financial Stability Signals from the IMF Country Report on South Korea (2022-2023)

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https://colab.research.google.com/drive/1ort6vyYidTpSBAtsf67JNQ9Qm4u8qxuX?usp=sharing

1. Introduction and method overview

Given how long and lengthy the IMF reports are, using language pipelines would be the best idea, The project turns the IMF report on South korea into outputs that can be checked: section summaries, theme counts, a JSON table of core indicators, grounded answers to factual questions, and a four panel dashboard. Everything runs in one Google Colab notebook.

The pipeline has 5 stages,

Firstly the pypdf extracts text of 87 pages. Lines that repeat on at least 30% of the pages including the headers and footers, and bare the page numbers are excluded automatically, Hyphenated line breaks are joined and all caps lines are used as section heading. In total there are 8 section and 27500 words.

Secondly, the text is cut into 113 chunks of 400 tokens with a 50 token overlap, so that context is maintained.

Thirdly, chunks are stored in a FAISS `IndexFlatL2` index.

Fourth, Gemini 3.8 Flash writes and extracts, and a different model, Gemini 3.5 Flash, acts as the judge. And finally, all the outputs are saved as JSON and plotted.

Every model answer is cached on Google Drive as soon as it arrives, with temperature 0, so the notebook can be re-run without repeating calls. I had a free tier limit of 5 requests per minute and 20 per day on the main models, which shaped the design: batching, caching and a 13 second pause between calls.

2. Summarisation and prompt quality

Prompts. The prompts are short and strict. These are the key wordings.

> Summariser (system): "Use only the text you are given. Never add numbers or facts that are not in the text."
> User prompt: "Summarise this section of an IMF report in fewer than 100 words, as one plain paragraph. Keep the main finding, the key numbers and any risk judgement."
>
> Judge: "Score the SUMMARY against the SOURCE from 1 to 5 on consistency, completeness and factual alignment. First list up to 3 important points from the SOURCE that the summary leaves out. Be strict: give 5 only when nothing important is missing. Return JSON only."

I made sure the word limit is checked, as we know models are not that good with counting words, so 100 or more words are sent back once to be shortened and the code stops with an error if it is still too long, and the judge reply is validated from 1 to 5

The judge prompt changed after my first run. In the first version the judge was the same lite model as the writer and gave 5 out of 5 on every dimension for every section, which told me nothing. Asking it to list what is missing before scoring, and using a different and stronger model, produced scores that vary.

Section	Source words	Summary words	Completeness first pass to final	Revised
Introduction	285	90	5 to 5	no
For Immediate Release	2,036	93	3 to 5	yes
Recent Economic Developments	1,498	92	4 to 4	no
Outlook and Risks	1,417	95	4 to 4	no
Policy Discussion	8,259	93	3 to 4	yes

Section	Source words	Summary words	Completeness first pass to final	Revised
Staff Appraisal	9,004	94	4 to 4	no
FSC/FSS	961	85	3 to 3	yes
Statistical Issues	4,068	89	3 to 4	yes

All summaries stayed under 100 words. Consistency and factual alignment averaged 5.0, while completeness averaged 3.6 on the first pass and 4.1 after revision. The revision loop therefore did its job in three of four cases (3 to 5, 3 to 4, 3 to 4), and in the FSC/FSS section it did not help (3 to 3). I saw a clear pattern, the summaries were accurate but not complete, I believe it's the 100 words limit. The two largest sections hold about 63% of the report, so each is compressed by roughly 90 to 1. The Staff Appraisal is also much longer than I would expect, which suggests it may have absorbed later parts of the report because heading detection missed their titles. No section was longer than the 150,000 characters limit, so nothing was cut off before summarising.

The exec summary was constructed from the 8 section summaries and contains 7 bullets. The key points are: a sound financial system with robust banks but a requirement to tighten the borrower based limits; very high levels of household debt and property prices which are in overvaluation territory; downside risks to growth from global monetary normalisation, geopolitics and Omicron; the policy rate having been raised by 75bps from 0.5%; public debt is projected to be at 60% of GDP by 2027; and the official view that the risks to households are manageable. The judge scored it 5, 4 and 5 and noted three omissions: the 2022% growth forecast for 2022, the Korean New Deal structural reforms and the climate recommendations on carbon pricing. My first exec summary had a 3.7% inflation figure in it which the judge said wasn't in the source. That was because the judge had been given a different bit of the text to the writers so I've amended the code to ensure the judge gets the same sections as the writers.

3. Quality of the structured outputs

For theme Distribution I used 2 methods, Method one counts keyword matches over the whole text with a regular expression list per theme. Method two sends every chunk to Gemini 3.5 Flash-Lite in batches of 8 and asks which of the 7 themes (0 to 3 per chunk) it substantially discusses. Flash-Lite is neither the writer nor the judge. The main measure is the number of chunks that discuss a theme, and shares are shares of theme assignments, not separate parts of the report, because one chunk can carry several themes. The model returned labels for 105 of the 113 chunks; one batch of 8 did not come back, so those chunks are unclassified.

Theme	Keyword chunks	Gemini chunks	Gemini share	Gemini only	Keyword only
Inflation & Prices	69	40	19.9%	4	33
Interest Rates & Credit	57	33	16.4%	2	26
Housing & Real Estate	41	33	16.4%	2	10
Sovereign Debt	57	29	14.4%	8	36
Systemic Financial Risk	42	28	13.9%	11	25
Labor Market Dynamics	32	23	11.4%	1	10
Banking Capital & Liquidity	37	15	7.5%	3	25

The most recurring is inflation and the least is banking capital. Keyword counting everyday words. "Fiscal" and "bond" push Sovereign Debt to 57 chunks against 29 for Gemini, and "bank" gives Banking Capital 37 against 15. Housing moves up from fifth place by keywords to joint second by Gemini, which suggests that housing risks are discussed in many chunks without the narrow keywords. Systemic Financial Risk has 11 chunks that only Gemini assigned, which fits the point that macroprudential concerns are often described indirectly. I checked a random sample of 5 classified chunks by eye. Three looked right, but two were doubtful: a statistical table chunk was labelled Sovereign Debt, Inflation and Interest Rates, and a growth

policy passage was labelled Labor Market and Sovereign Debt. So the classifier is useful, but it is noisy on tables, and I did not measure its precision.

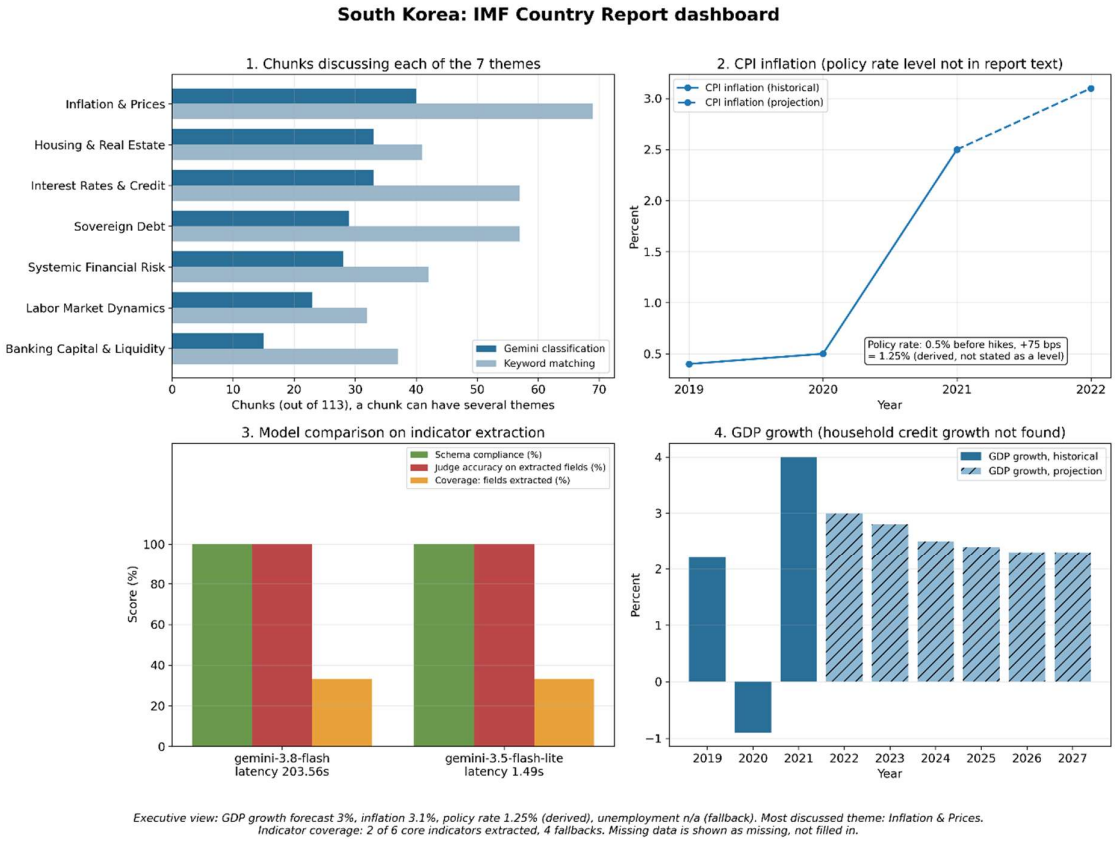
5. Extension: model comparison

I compared Gemini 3.8 Flash and Gemini 3.5 Flash-Lite on the same extraction task, with the same context and prompt and model switching switched off.

Model	Schema compliance	Latency (s)	Fields extracted	Judge accuracy
gemini-3.8-flash	100%	203.6	2 of 6	100%
gemini-3.5-flash-lite	100%	1.5	2 of 6	100%

Both models produced valid JSON and found the same two fields, so on quality they tied. The only difference was speed, and it was large: Flash-Lite answered in about 1.5 seconds against more than 200 seconds for 3.8 Flash. This is a single run on a shared free tier, and the first run also hit overload errors, so the latency gap says more about server load than about the models. The sensible conclusion is that for a narrow extraction task with a strict schema, the cheaper model is enough, and that the bottleneck is retrieval and the report's own content, not model size.

6. Dashboard and analytical discussion



Themes (panel 1). Inflation and prices dominate the report on both measures, followed by interest rates and housing. The gap between the dark and light bars shows how much keyword counting overstates sovereign debt and banking.

Inflation (panel 2). CPI inflation was 0.4% in 2019 and 0.5% in 2020, then rose to 2.5% in 2021 with 3.1% projected for 2022. The speed of this change, from near zero to above 3% in two years, explains why

inflation is the leading theme. The report expects core inflation to stay in the mid 2 percent range in 2022 and fall to about 2 percent in 2023 as supply pressures fade. A policy rate line is missing because the report gives only the size of the hikes. The note on the panel shows the derived 1.25%, clearly labelled. I did not invent yearly rate values to fill the gap.

Growth (panel 4). Real GDP growth was 2.2% in 2019, fell to -0.9% in 2020, rebounded to 4.0% in 2021 and is projected at 3.0%, 2.8%, 2.5%, 2.4% and 2.3% from 2022 to 2026, then 2.3% in 2027. The recovery was fast, but the projections show a steady fade toward lower trend growth, which the report links to Korea's ageing population. The household credit growth series was not found in the text, so the link between credit and tightening is not tested here.

Implications. The main risk chain in the report runs from higher global and domestic rates to households. Household debt is very high, a large part carries floating rates, and many loans are repaid as one lump sum at the end, with Jeonse deposits adding rollover risk when house prices turn. Banks look well protected, with low NPL ratios described in the text and a stress test finding of resilience, so the risk is mainly to borrowers and demand rather than to bank capital. The IMF's response is tighter borrower-based measures, a sectoral countercyclical capital buffer for household exposures, a phased withdrawal of support to small firms, and a rules-based fiscal framework, given that public debt is projected to reach 60 percent of GDP by 2027.